

# MIT OpenCourseWare

14.02SC

Principles of Macroeconomics 14.02SC — Principles of Macroeconomics May 2026

**EXTREMELY HARD – MIT LEVEL****Time Limit: 3 hours****Total Marks: 100**

1. (3 points) Construct an original proof-level problem involving GDP for 14.02SC (Principles of Macroeconomics) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.
2. (5 points) Analyze the limitations of standard approaches to Inflation when applied to edge cases in 14.02SC. Construct explicit counterexamples showing where intuitive reasoning fails.
3. (9 points) Analyze the limitations of standard approaches to Unemployment when applied to edge cases in 14.02SC. Construct explicit counterexamples showing where intuitive reasoning fails.
4. (3 points) Prove the fundamental theorem concerning Fiscal Policy in the context of 14.02SC (Principles of Macroeconomics). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
5. (5 points) Prove the fundamental theorem concerning Monetary Policy in the context of 14.02SC (Principles of Macroeconomics). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
6. (7 points) Analyze the limitations of standard approaches to Economic Growth when applied to edge cases in 14.02SC. Construct explicit counterexamples showing where intuitive reasoning fails.
7. (3 points) Analyze the limitations of standard approaches to GDP when applied to edge cases in 14.02SC. Construct explicit counterexamples showing where intuitive reasoning fails.
8. (4 points) Analyze the limitations of standard approaches to Inflation when applied to edge cases in 14.02SC. Construct explicit counterexamples showing where intuitive reasoning fails.

9. (5 points) Prove the fundamental theorem concerning Unemployment in the context of 14.02SC (Principles of Macroeconomics). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
10. (3 points) Construct an original proof-level problem involving Fiscal Policy for 14.02SC (Principles of Macroeconomics) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.
11. (2 points) Construct an original proof-level problem involving Monetary Policy for 14.02SC (Principles of Macroeconomics) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.
12. (4 points) Construct an original proof-level problem involving Economic Growth for 14.02SC (Principles of Macroeconomics) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.
13. (5 points) Construct an original proof-level problem involving GDP for 14.02SC (Principles of Macroeconomics) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.
14. (2 points) Construct an original proof-level problem involving Inflation for 14.02SC (Principles of Macroeconomics) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.
15. (3 points) Prove the fundamental theorem concerning Unemployment in the context of 14.02SC (Principles of Macroeconomics). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
16. (6 points) Analyze the limitations of standard approaches to Fiscal Policy when applied to edge cases in 14.02SC. Construct explicit counterexamples showing where intuitive reasoning fails.
17. (4 points) Prove the fundamental theorem concerning Monetary Policy in the context of 14.02SC (Principles of Macroeconomics). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
18. (3 points) Construct an original proof-level problem involving Economic Growth for 14.02SC (Principles of Macroeconomics) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.
19. (4 points) Prove the fundamental theorem concerning GDP in the context of 14.02SC (Principles of Macroeconomics). State the theorem precisely, provide a complete proof, and discuss three important corollaries.

20. (2 points) Prove the fundamental theorem concerning Inflation in the context of 14.02SC (Principles of Macroeconomics). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
21. (4 points) Construct an original proof-level problem involving Unemployment for 14.02SC (Principles of Macroeconomics) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.
22. (3 points) Prove the fundamental theorem concerning Fiscal Policy in the context of 14.02SC (Principles of Macroeconomics). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
23. (4 points) Analyze the limitations of standard approaches to Monetary Policy when applied to edge cases in 14.02SC. Construct explicit counterexamples showing where intuitive reasoning fails.
24. (4 points) Analyze the limitations of standard approaches to Economic Growth when applied to edge cases in 14.02SC. Construct explicit counterexamples showing where intuitive reasoning fails.
25. (3 points) Prove the fundamental theorem concerning GDP in the context of 14.02SC (Principles of Macroeconomics). State the theorem precisely, provide a complete proof, and discuss three important corollaries.

## Solutions

### **Solution 1:**

Solution approach: First recall the definitions and key theorems related to GDP from 14.02SC. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

### **Solution 2:**

The edge case analysis reveals the subtle assumptions underlying standard treatments of Inflation. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

### **Solution 3:**

The edge case analysis reveals the subtle assumptions underlying standard treatments of Unemployment. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

### **Solution 4:**

The proof follows from the definitions and properties established in 14.02SC. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

### **Solution 5:**

The proof follows from the definitions and properties established in 14.02SC. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

### **Solution 6:**

The edge case analysis reveals the subtle assumptions underlying standard treatments of Economic Growth. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

### **Solution 7:**

The edge case analysis reveals the subtle assumptions underlying standard treatments of GDP. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

### **Solution 8:**

The edge case analysis reveals the subtle assumptions underlying standard treatments of Inflation. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

### **Solution 9:**

The proof follows from the definitions and properties established in 14.02SC. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

### **Solution 10:**

Solution approach: First recall the definitions and key theorems related to Fiscal Policy from 14.02SC. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

**Solution 11:**

Solution approach: First recall the definitions and key theorems related to Monetary Policy from 14.02SC. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

**Solution 12:**

Solution approach: First recall the definitions and key theorems related to Economic Growth from 14.02SC. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

**Solution 13:**

Solution approach: First recall the definitions and key theorems related to GDP from 14.02SC. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

**Solution 14:**

Solution approach: First recall the definitions and key theorems related to Inflation from 14.02SC. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

**Solution 15:**

The proof follows from the definitions and properties established in 14.02SC. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

**Solution 16:**

The edge case analysis reveals the subtle assumptions underlying standard treatments of Fiscal Policy. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

**Solution 17:**

The proof follows from the definitions and properties established in 14.02SC. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

**Solution 18:**

Solution approach: First recall the definitions and key theorems related to Economic Growth from 14.02SC. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

**Solution 19:**

The proof follows from the definitions and properties established in 14.02SC. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

**Solution 20:**

The proof follows from the definitions and properties established in 14.02SC. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

**Solution 21:**

Solution approach: First recall the definitions and key theorems related to Unemployment from 14.02SC. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

**Solution 22:**

The proof follows from the definitions and properties established in 14.02SC. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

**Solution 23:**

The edge case analysis reveals the subtle assumptions underlying standard treatments of Monetary Policy. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

**Solution 24:**

The edge case analysis reveals the subtle assumptions underlying standard treatments of Economic Growth. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

**Solution 25:**

The proof follows from the definitions and properties established in 14.02SC. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.